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Quantum Computing

Overview of Quantum Computing. How does it compare to the conventional Von Neumann architecture?

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DECLARATION

I hereby declare that this report is entirely my own work. All ideas, content, and findings presented in this report are the result of my independent research and analysis. Any external sources referenced in the report are duly credited and acknowledged.

I affirm that I have not plagiarized any part of this report, and all information obtained from external sources is properly cited.

I understand the consequences of academic dishonesty and affirm that this report represents my genuine effort and original contribution to the understanding of the subject matter.

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ABSTRACT

This report is all about Quantum Computing and how it's different from the computers we use. We'll start with a quick look back at the history of computers and then get into the basics of Quantum Computing, explaining things like quantum bits or qubits. We'll also talk about the special parts of a Quantum Computer and how they work. After that, we'll compare Quantum Computing with the computers we're used to, like the one you might be using right now. We'll discuss how Quantum Computers can be way faster for some jobs and do things in a super cool, parallel way.

The report also touches on where we might use Quantum Computing, like making secret codes, solving tricky problems, and teaching computers to learn. But it's not all rainbows and sunshine; we'll talk about the challenges Quantum Computers face, like trying to stay accurate and not making mistakes.

KEYWORDS

1. Quantum Computing:

   Quantum computing is a field of computing that leverages principles of quantum mechanics to perform computations. Unlike classical computers, which use bits to represent 0s and 1s, quantum computers use qubits, which can exist in a superposition of states, allowing them to represent both 0 and 1 simultaneously. Quantum computing has the potential to solve certain problems much faster than classical computers.

2. Von Neumann architecture:

   The Von Neumann architecture is the traditional computer architecture proposed by mathematician and physicist John von Neumann. It consists of a central processing unit (CPU), memory, control unit, arithmetic logic unit (ALU), and input/output devices. This architecture follows a sequential processing model, where instructions are executed one at a time, and data and instructions share the same memory.

3. Qubits:

   Qubits are the basic units of information in quantum computing. Unlike classical bits, which can be in one of two states (0 or 1), qubits can exist in a superposition of both 0 and 1 simultaneously. This unique property enables quantum computers to perform parallel computations and solve certain problems more efficiently than classical computers.

4. Conventional Computing:

   Conventional computing, also known as classical computing, refers to the traditional model of computing that follows classical physics principles. It uses classical bits to represent information in binary form (0s and 1s) and follows the Von Neumann architecture. Conventional computers are the everyday devices we use, such as personal computers, laptops, and smartphones.

5. Classical Bits:

   Classical bits are the basic units of information in classical computing. They can exist in one of two states: 0 or 1. Classical bits are the building blocks of classical computers, and information is processed using logical operations based on these binary states. In contrast to qubits in quantum computing, classical bits do not have the ability to exist in a superposition of states.

ACKNOWLEDGEMENT

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1. INTRODUCTION

1.1. History of Computing

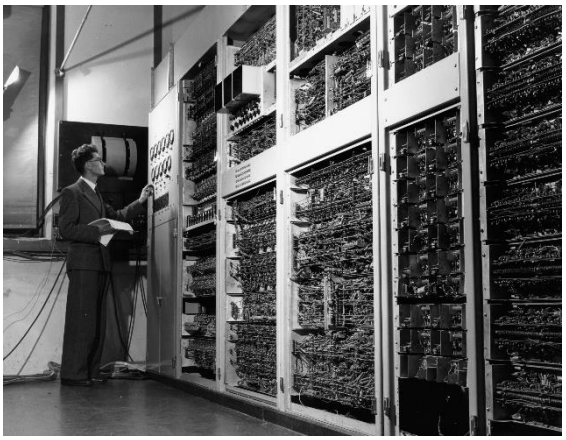


Figure 1 - Computer History - 1946

Computers have a fascinating story that goes back many years. It all began with simple tools like the abacus, which people used in the 17th century to do basic math. Then, in the 19th century, smart minds like Charles Babbage and Ada Lovelace had big ideas about creating a computer that could follow instructions. Jumping to the 20th century, we got electronic computers, thanks to people like Alan Turing. They made cool stuff like the ENIAC and the integrated circuit, which are like the building blocks of today's computers. All these inventions and discoveries have made our computers today super powerful and part of our everyday lives. (O'regan sp, london)

- Abacus Origins (17th Century): People started with tools like the abacus to do simple math.
- 19th Century Visionaries: Charles Babbage and Ada Lovelace had big ideas about creating a computer that could follow instructions.
- 20th Century Breakthroughs: Alan Turing and others made electronic computers, like the ENIAC, and discovered the integrated circuit.
- Modern Computers: All these inventions laid the foundation for the powerful computers we use today, making our lives more connected and convenient.

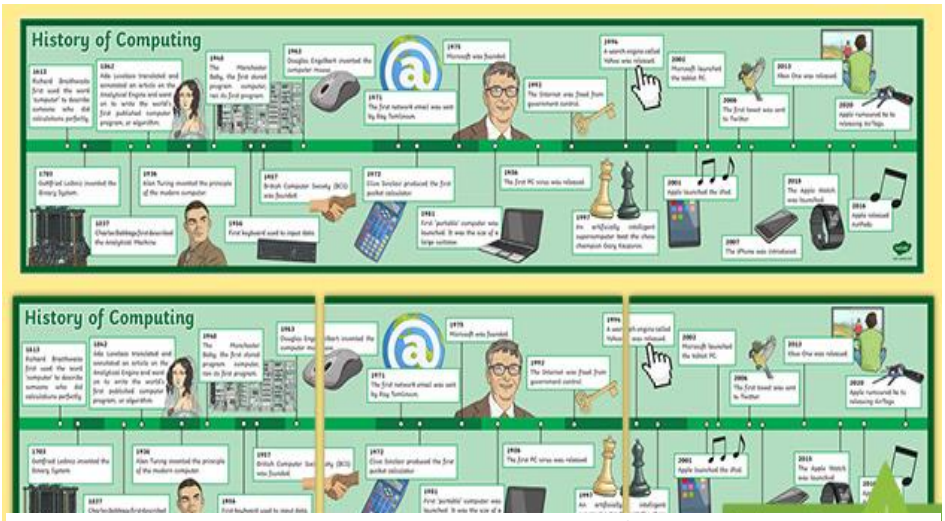


Figure 2 - Computer History - Timeline

1.2. Introduction to Quantum Computing

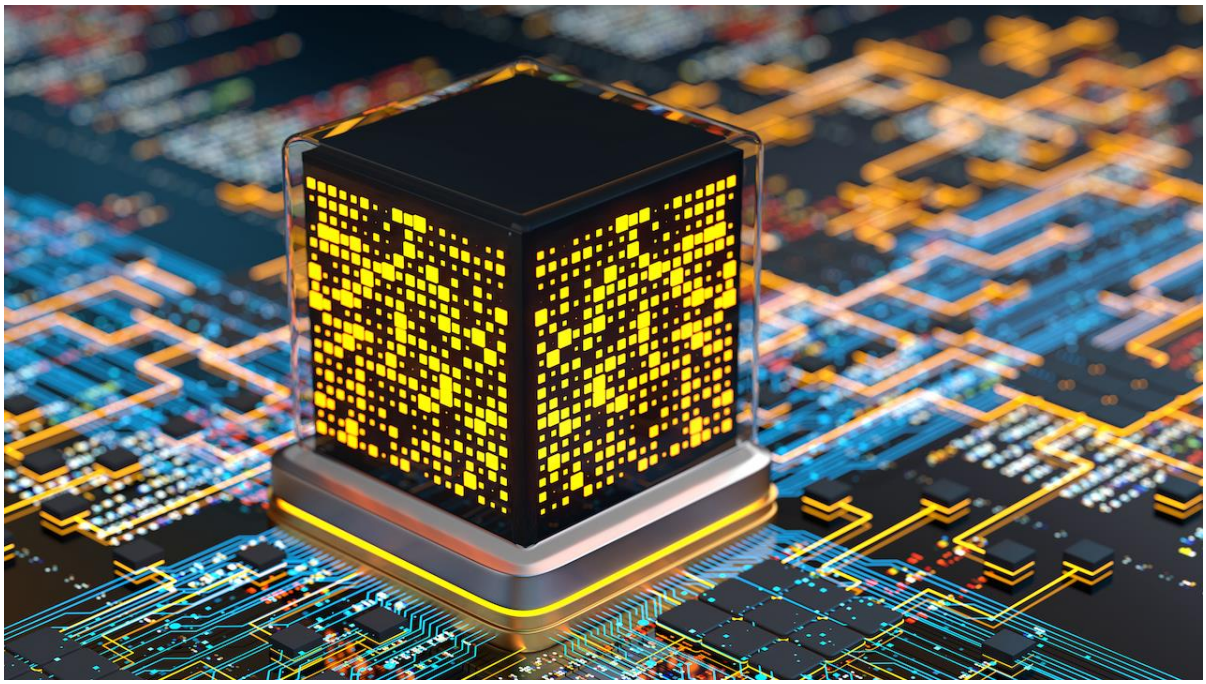


Figure 3 - Quantum Computing

Quantum Computing emerges as a transformative paradigm, challenging the conventions of classical computation. Unlike traditional computers that rely on bits with a binary state (zero or one), Quantum Computers rely on qubits. These quantum bits or qubits possess the unique property of superposition, enabling them to exist in multiple states.

As we delve into the intricacies of Quantum Computing, it becomes evident that this technology has the capacity to address complex problems with unprecedented efficiency. (Hughes et al.)

2. QUANTUM COMPUTING VS. CONVENTIONAL VON NEUMANN COMPUTER ARCHITECTURE

	Conventional Computing	Quantum Computing
Memory & Storage	Separates memory and processing.	Allows both storage and processing together.
Data Speed	May have bottlenecks in speed.	Potential to be faster, overcoming limits.
Error Correction	Uses established techniques.	Research Ongoing.
Information Processing	Follows predictable rules.	Bringing new possibilities.

Table 1 - Quantum computing VS Conventional Computing

2.1. Memory and Storage

Conventional Computing:

Traditional computers keep memory (RAM) and processing (CPU) separate. This setup can cause slowdowns in data transfer and processing speed.

Quantum Computing:

Quantum computers do things differently, letting information be both stored and processed at the same time. This new way of doing things has the potential to overcome the limits of regular computers.

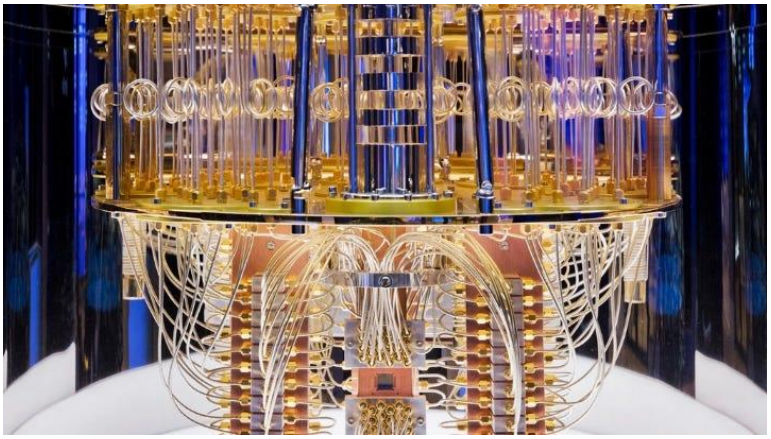


Figure 4 - Navigating the challenges

2.2. Speed and Efficiency

Conventional Computing:

The sequential nature of processing in Von Neumann architecture imposes limitations on speed, often referred to as the von Neumann bottleneck. This hinders the overall efficiency, especially when handling complex problems that demand rapid computation.

Quantum Computing:

Quantum computing, on the other hand, presents a paradigm shift by offering the potential to solve certain problems exponentially faster.

2.3. Error Correction

Conventional Computing:

Classical computing has established error correction techniques to ensure the accuracy and reliability of computations.

Quantum Computing:

Quantum computing introduces unique challenges related errors. Ongoing research focuses on developing quantum error correction algorithms to effectively mitigate these challenges.

2.4. Information Processing

Conventional Computing:

Follows deterministic principles, where each operation produces predictable outcomes.

Quantum Computing:

The final result of a quantum computation is determined by the probabilities associated with different states.

CONCLUSION

In conclusion, our exploration of computing evolution and the emergence of Quantum Computing underscores a pivotal moment in technological advancement. The contrast between conventional Von Neumann architecture and the innovative principles of Quantum Computing showcases the latter's potential to redefine computational possibilities.

While Quantum Computing introduces unprecedented speed, simultaneous information processing, and promising advancements, challenges like classical bottlenecks and quantum uncertainties persist. The way forward lies in a harmonious collaboration between classical and quantum systems, offering a synergy that transcends current limitations.

As we navigate this dynamic intersection of computing paradigms, the convergence of classical and quantum elements charts a course toward a future where the boundaries of computational capabilities are redefined. The report stands as a testament to the transformative potential of Quantum Computing, inviting further exploration and collaboration in the ever-evolving landscape of information technology.

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